

① Line



2) Polyline



3) Circle



4) Arc



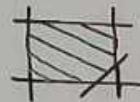
5) Rectangle



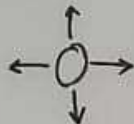
6) Ellipse



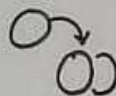
7) Hatch



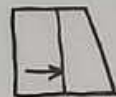
8) Move



9) Copy



10) Stretch



11) Rotate



12) Mirror



13) Scale



14) Trim



15) Fillet



16) Array



Auto Cad Symbols:-

17) Erase



18) Explode



19) Offset



Sketch free hand the graphical symbols for the following Engineering items.

Electronics and Electrical

Amplifier		Antenna		Arrester		Battery	
Circuit Breaker		Capacitor		Coil		Connector	
Electric Contact		Core		Fuse		Directional Coupler	
Ground		Handset		Rectifier		Visual Signaling device	
Receiver		Repeater		Ground		Handset	
Thermocouple		Inductor Winding		Incandescent Lamp		Ballast Lamp	
Transformer		Switch		Resistor		Transmission Path	
Generator		Motor		Solenoid		Thermostat	
Thermistor		Variable Resistor		Voltmeter		NPN-type Transistor	
PNP-type Transistor		Direct Current		Alternating Current		Power Frequency	
Apparatus & Machine suitable for DC or AC		Half-wave Rectifier		Line or Cable existing		Line or Cable planned	
Silicon Controlled Rectifier		Power Line		Underground Cable		Overhead Line	

Electronics and Electrical

Conductors or a group of Conductors		Flexible Conductors		Two Conductors		Three Conductors	
Four Conductors		Junction of Conductors		Crossing without Electrical Connection		Crossing and Connecting Conductors	
Frame and Chassis Connection		Fault		Fault to Frame		Earth Fault	
Mechanically Coupled Machine		DC Generator		AC Generator		DC Motor	
AC Motor Single Phase		Three Phase Motor		Three Phase Motor in Delta Connection		One Way Switch (Single Pole Switch)	
Two Pole Switch		Three Pole Switch		Two Way Switch		Intermediate Switch	
Push Button Switch		Socket Outlets		Socket Outlets, 5A		Socket Outlets, 5A with Switch	
Socket Outlets, 15A with Switch		Lamp mounted on ceiling		Group of 3, 40 Watt Lamp		Lamp mounted on wall	
Fluorescent Lamp		Ceiling Fan		Wall mounted Fan		Exhaust Fan	
Fan Regulator		Bell		Pickup		Buzzer	
Siren		Public addressing System		Diode with Filament		Telephone-Telegraph Line	

Electronics and Electrical

Conductors or a group of Conductors		Flexible Conductors		Two Conductors		Three Conductors	
Four Conductors		Junction of Conductors		Crossing without Electrical Connection		Crossing and Connecting Conductors	
Frame and Chassis Connection		Fault		Fault to Frame		Earth Fault	
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Siren		Public addressing System		Diode with Filament		Telephone-Telegraph Line	

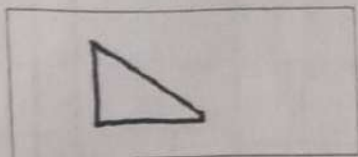
the graphical symbols for the following Engineering items.

Electronics and Electrical

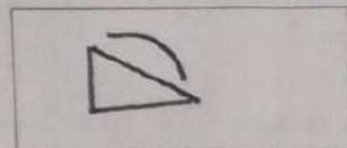
Amplifier		Antenna		Arrester		Battery	
Circuit Breaker		Capacitor		Coil		Connector	
Electric Contact		Core		Fuse		Directional Coupler	
Ground		Handset		Rectifier		Visual Signaling device	
Receiver		Repeater		Ground		Handset	
Thermocouple		Inductor Winding		Incandescent Lamp		Ballast Lamp	
Transformer		Switch		Resistor		Transmission Path	
Generator		Motor		Solenoid		Thermostat	
Thermistor		Variable Resistor		Voltmeter		NPN-type Transistor	
PNP-type Transistor		Direct Current		Alternating Current		Power Frequency	
Apparatus & Machine suitable for DC or AC		Half-wave Rectifier		Line or Cable existing		Line or Cable planned	
Controlled Rectifier		Power Line		Underground Cable		Overhead Line	

Sketch freehand the graphical symbols for the following welding items.

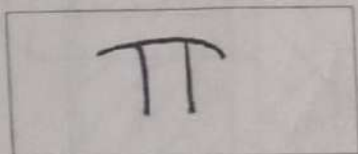
Lap Weld



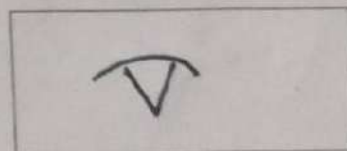
Fillet



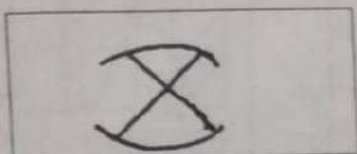
Square Butt



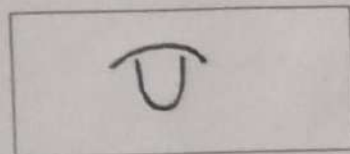
Single V-Butt



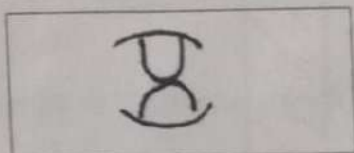
Double V-Butt



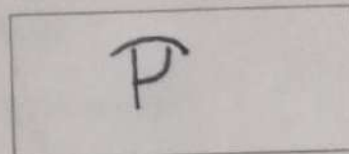
Single U-Butt



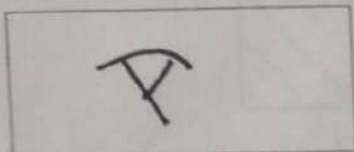
Double U- Butt



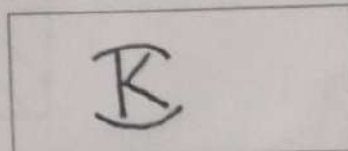
Single J-Butt



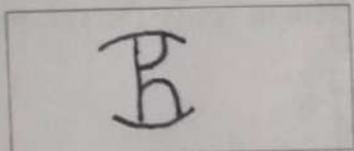
Single Bevel Butt



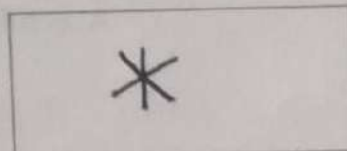
Double Bevel Butt



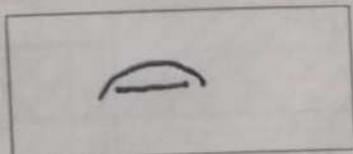
Double J-Butt



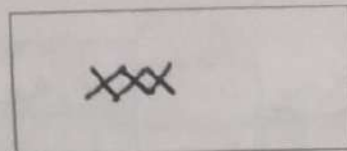
Spot Weld



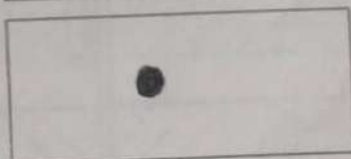
Bead or Edge Weld



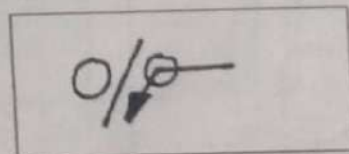
Seam Weld



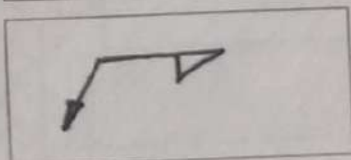
Field Weld



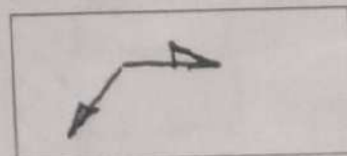
Weld all around



Fillet Weld on own side of joints



Fillet weld on opposite side of joint



Fillet weld on both sides of joint

